

RESEARCH ARTICLE Hot Paper

Surface Oxide Species on Metal Cocatalysts Mediate Pathways in Water-Oxidation-Coupled CO₂ Photoreduction: Insights From Pd

Bifang Li¹ | Ziyi Sun¹ | Bo Su¹ | Xiahui Lin² | Wandong Xing¹ | Yuanfu Ren³ | Kunlong Liu¹ | Yidong Hou¹  |
Xue Feng Lu¹  | Lihua Lin¹ | Masakazu Anpo¹ | Huabin Zhang³  | Sibowang Wang¹ 

¹State Key Laboratory of Chemistry for NBC Hazards Protection, State Key Laboratory of Photocatalysis on Energy and Environment, College of Chemistry, Fuzhou University, Fuzhou, China | ²College of Environment and Safety Engineering, Fuzhou University, Fuzhou, China | ³Center For Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology (KAUST), Thuwal, Kingdom of Saudi Arabia

Correspondence: Huabin Zhang (huabin.zhang@kaust.edu.sa) | Sibowang Wang (sibowang@fzu.edu.cn)

Received: 9 March 2026 | **Revised:** 13 April 2026 | **Accepted:** 17 April 2026

Keywords: CO₂ reduction | cocatalyst engineering | photocatalysis | surface chemistry

ABSTRACT

Photocatalytic overall CO₂ reduction using H₂O as an electron donor is hindered by sluggish reaction kinetics and poorly defined interfacial pathways. Here, we demonstrate that surface oxide species on metal cocatalysts act as intrinsic mediators of the reaction pathway in water-oxidation-coupled CO₂ photoreduction. Taking Pd as a model system, we construct a Pd@PdO_x cocatalyst on TiN, in which metal Pd is partially encapsulated by spontaneously formed surface PdO_x species. This pre-formed Pd/PdO_x interface establishes a bifunctional reaction landscape that spatially coordinates proton management and CO₂ activation. In situ spectroscopic analyses combined with theoretical calculations reveal that PdO_x domains preferentially adsorb and activate CO₂, while adjacent metal Pd sites function as proton reservoirs derived from water oxidation. Directional proton transfer across the Pd/PdO_x interface lowers the barrier for *COOH formation, suppresses the competing H₂ evolution reaction, and promotes selective CO release. Under full-spectrum irradiation, Pd@PdO_x/TiN achieves a CO yield rate of 200 μmol g⁻¹ h⁻¹ with 81% selectivity, substantially outperforming the counterparts dominated by either Pd or PdO_x. This study highlights surface oxide species as structural determinants of pathway selection and provides mechanistic insights for engineering metal cocatalysts for efficient and selective CO₂ photoreduction.

1 | Introduction

Artificial photosynthesis harnesses solar energy to couple CO₂ reduction and H₂O oxidation, providing a sustainable route for producing carbonaceous fuels while addressing global energy and environmental challenges [1–4]. Extensive pioneering studies on CO₂ photoreduction with pure water have provided valuable insights into catalyst design, band structure engineering, and

cocatalyst modification to enhance activity and selectivity [5–12]. Despite these progresses, the efficiency of photocatalytic overall CO₂ reduction remains unsatisfactory. This limitation arises from the thermodynamic stability of CO₂ molecules, sluggish multi-electron/proton transfer kinetics, and high activation barriers associated with key reaction intermediates [13–16]. Therefore, augmenting CO₂ adsorption and activation via rational surface engineering [17, 18], optimizing proton-coupled electron

migration, and facilitating the formation of the key $^*\text{COOH}$ intermediate are crucial for achieving efficient and selective CO_2 -to-CO conversion.

Noble metals have been extensively applied as cocatalysts to accelerate CO_2 photoreduction, with recent studies emphasizing the importance of cooperative interactions between distinct metal states, such as nanoparticles (NPs) and single atoms [19–22]. However, surface oxidation of metals—inevitable under ambient exposure or catalytic conditions—is typically regarded as incidental rather than functional, and its mechanistic role in pathway regulation remains underexplored [23–25]. Taking palladium (Pd) as a representative, metal Pd can efficiently collect $^*\text{H}$ species owing to favorable hydrogen binding, but its filled $4d^{10}$ configuration restricts its electron-accepting ability [26–28]. The accumulated $^*\text{H}$ species may recombine to form H_2 or participate in proton-coupled CO_2 reduction. In contrast, oxidized Pd (PdO_x) species with unoccupied d orbitals can facilitate electron acceptance and enable d – π interactions with CO_2 , weakening C=O bonds and lowering CO_2 activation barriers [29–31]. These complementary properties suggest that intimate Pd/ PdO_x coupling could create a bifunctional interface, where Pd enriches reactive $^*\text{H}$ species while adjacent PdO_x activates CO_2 , enabling interfacial proton transfer and selective $^*\text{COOH}$ formation.

Motivated by this hypothesis, we have deliberately designed a Pd@PdO_x cocatalyst supported on TiN, a metallic photocatalyst with nearly full-spectrum absorption [32]. The Pd@PdO_x structure forms spontaneously during cocatalyst deposition, generating a stable metal/oxide interface. Under illumination, electrons excited in TiN transfer to Pd@PdO_x across the Schottky junction, while holes remain on TiN to drive water oxidation. Within the cocatalyst, PdO_x domains preferentially adsorb and activate CO_2 , whereas adjacent Pd domains function as proton collectors that shuttle $^*\text{H}$ species toward PdO_x sites. This directional proton transfer lowers the barrier for $^*\text{COOH}$ formation, suppresses the competing H_2 evolution reaction, and promotes selective CO release. Using H_2O as the electron donor, $\text{Pd@PdO}_x/\text{TiN}$ affords a CO production rate of $200\ \mu\text{mol g}^{-1}\ \text{h}^{-1}$ with 81% selectivity, markedly surpassing the control catalysts enriched by either Pd or PdO_x . These findings reveal that surface oxide species on metal cocatalysts—often overlooked yet ubiquitously present—act as structural determinants of interfacial pathway selection, offering a general design principle for efficient and selective solar-driven CO_2 photoreduction.

2 | Results and Discussion

A facile synthesis method leveraging the intrinsic surface oxidation tendency of Pd was developed to deposit core-shell Pd@PdO_x NPs onto TiN, yielding the $\text{Pd@PdO}_x/\text{TiN}$ catalyst (Figure 1a). In this process, Pd^{2+} ions are first adsorbed onto the TiN surface via impregnation. Subsequently, O_2 bubbling followed by NaBH_4 addition initiates a dynamic redox interplay: Pd^{2+} species are rapidly reduced to metal Pd^0 by NaBH_4 , while the freshly formed Pd^0 undergoes partial surface oxidation by dissolved O_2 . This synchronized reduction–oxidation sequence permits the formation of Pd@PdO_x cocatalyst comprising a Pd core wrapped by a PdO_x shell, immobilized on the TiN support.

Pristine TiN was synthesized via a nitridation route with TiO_2 nanosheets as the precursor (see Supporting Information for the details) [32, 33]. Powder X-ray diffraction (XRD) confirms the complete conversion of TiO_2 into cubic TiN, which adopts a radially contracted lamellar morphology (Figures S1 and S2). Elemental mappings reveal a uniform distribution of Ti and N throughout the TiN framework (Figure S2d). The metallic nature of TiN is verified by valence band X-ray photoelectron spectroscopy (VB-XPS) and temperature-dependent resistance tests (Figure S3a, b) [34–36]. TiN exhibits strong optical absorption spanning the ultraviolet to near-infrared regions, with an apparent optical band gap of ca. 1.22 eV and an electronic band structure well-suited for driving CO_2 reduction and H_2O oxidation (Figures S3c, d, and S4) [32, 37–39].

The as-prepared TiN was then dispersed in an aqueous PdCl_2 solution and impregnated for 1 h to enable the spontaneous adsorption of Pd^{2+} ions via electrostatic interactions (Figure S5). The adsorbed Pd^{2+} species were then subjected to a combined NaBH_4/O_2 treatment to obtain the 0.8-Pd@ PdO_x/TiN composite. By varying the amount of PdCl_2 precursor, two additional samples of 0.3-Pd@ PdO_x/TiN and 1.5-Pd@ PdO_x/TiN were synthesized using the same procedure, where the numerical values represent the theoretical Pd mass percentages. Inductively coupled plasma optical emission spectrometry (ICP-OES) confirms that the actual Pd contents closely match the theoretical values (Table S1). The 0.8-Pd@ PdO_x/TiN sample is selected for detailed studies and hereafter abbreviated as Pd@ PdO_x/TiN .

All Pd@ PdO_x/TiN materials exhibit only diffraction peaks corresponding to TiN (Figure S6), because of the low content and high dispersion of Pd species. Transmission electron microscopy (TEM) reveals that Pd@ PdO_x NPs are discretely distributed on the TiN surface with an average size of ca. 4.3 nm (Figure 1b). Aberration-corrected high-angle annular dark-field scanning TEM (AC-HAADF-STEM) reveals no discernible interfacial gap between Pd@ PdO_x and TiN (Figure 1c, d), indicating their intimate contact. The Pd@ PdO_x NPs present a unique core-shell structure, consisting of a crystalline core enveloped by a porous layer of ca. 1.05 nm in thickness. The lattice spacing of 0.23 nm observed in the core region corresponds to the (111) plane of Pd (Figure S7), whereas the shell lacks perceptible lattice fringes.

Three-dimensional atom-overlapping Gaussian fitting and fast Fourier transform (FFT) analyses show ordered bright spots and sharp diffraction spots in the core region (Figure 1f), in contrast to randomly distributed bright spots and diffuse diffraction rings in the shell (Figure 1e), validating the crystalline–amorphous structure of Pd@ PdO_x [40, 41]. HAADF-STEM-EDX elemental mapping displays uniform presence of Pd and obvious surface enrichment of O (Figure 1g). Consistently, electron energy-loss spectroscopy (EELS) unveils stronger Pd– M_3 and O–K edge signals in the shell region compared with the core (Figure 1h).

The surface chemical states of Pd@ PdO_x/TiN were analyzed by XPS. The Ti 2p spectrum exhibits three spin–orbit doublets assignable to Ti–N, N–Ti–O, and Ti–O bonds (Figure S8a) [42]. The appearance of Ti–O species originates from inevitable surface oxidation, a common feature of metal nitrides. The N 1s spectrum displays three peaks (Figure S8b), corresponding to N–Ti–O, N–Ti bonds, and a satellite feature, respectively

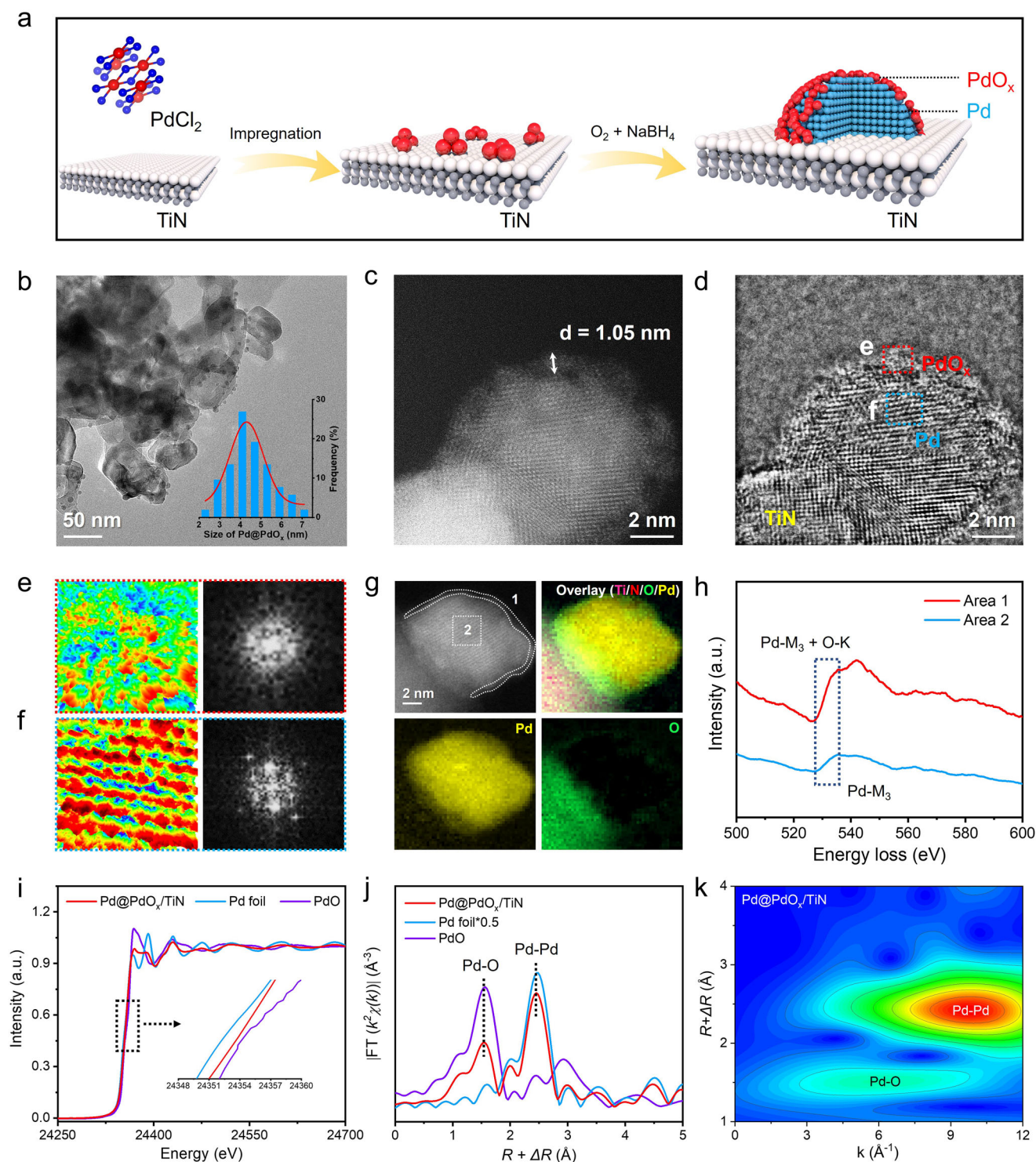


FIGURE 1 | (a) Schematic illustration of the synthesis process of Pd@PdO_x/TiN. (b) TEM image and corresponding particle size distribution of Pd@PdO_x. (c) AC-HAADF-STEM image, and (d) AC-IDPC image of Pd@PdO_x/TiN. (e, f) 3D atom-overlapping Gaussian-function fitting mappings and corresponding Fourier transform images of the red and blue wireframes in (d). (g) AC-HAADF-STEM image and elemental mappings of Pd@PdO_x/TiN. (h) EELS spectra of the corresponding Pd@PdO_x nanoparticle in (g). (i) XANES and (j) FT-EXAFS spectra of Pd@PdO_x/TiN, Pd foil, and PdO. (k) WT-EXAFS spectrum of Pd@PdO_x/TiN.

[43]. The high-resolution Pd 3d spectrum contains well-defined peaks of metal Pd⁰ (Figure S8c), accompanied by weaker signals assigned to Pd^{δ+} and Pd²⁺ species [44, 45]. The coexistence of Pd^{δ+} and Pd²⁺ reflects the presence of oxidized Pd species within the PdO_x shell. Quantitative analysis reveals that Pd⁰ accounts for

ca. 66% of the total Pd content, confirming the predominance of Pd⁰ with a smaller proportion of PdO_x. The above results validate the formation of Pd@PdO_x NPs composed of a Pd⁰ core encapsulated by a PdO_x shell, strongly anchored on TiN substrate.

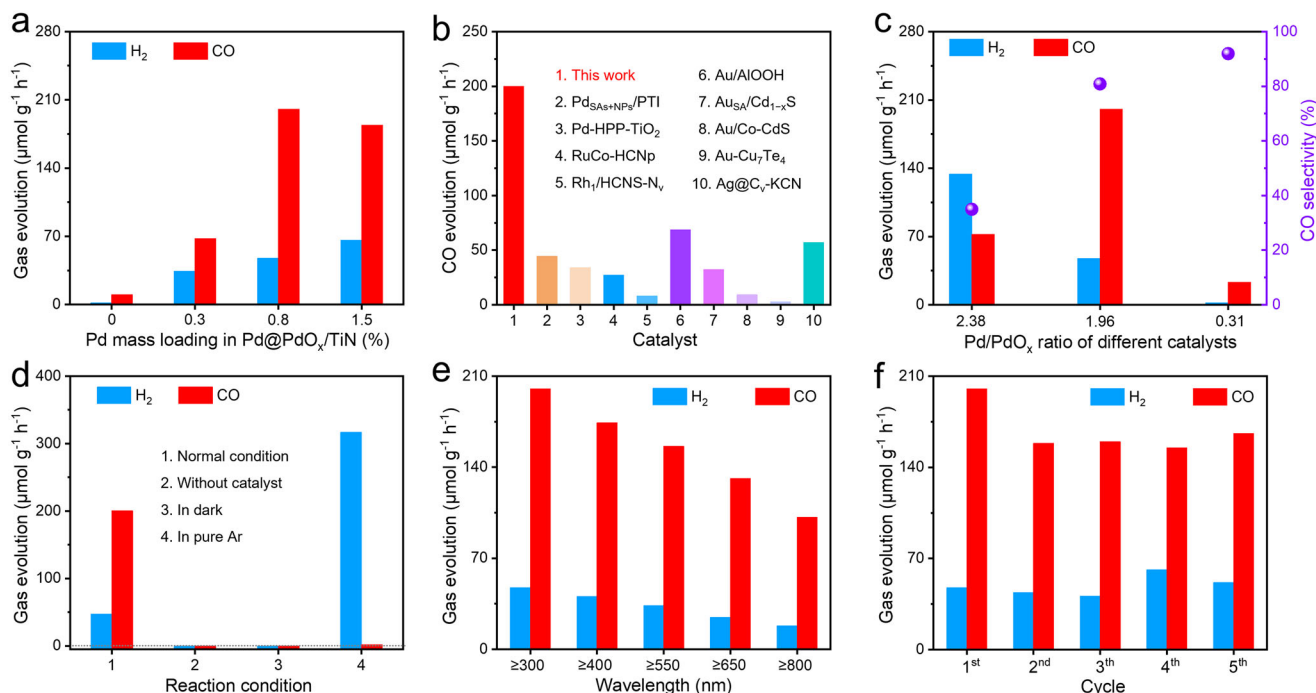


FIGURE 2 | (a) Photocatalytic CO₂ reduction performance of Pd@PdO_x/TiN catalysts with different Pd loadings. (b) Comparison of CO₂ photoreduction performance of Pd@PdO_x/TiN with reported catalysts. (c) CO₂ photoreduction performance of catalysts with different Pd/PdO_x ratios. (d) CO₂ reduction activities of the system under varied reaction conditions. (e) Wavelength-dependent photocatalytic performance measured at identical intensity and (f) Stability tests of Pd@PdO_x/TiN.

To gain insights into the coordination environment of Pd species, X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) analyses were conducted. In the Pd K-edge XANES spectra (Figure 1i), the absorption edge of Pd@PdO_x/TiN lies between those of Pd foil and PdO, indicating partial oxidation of Pd species with an average oxidation state between 0 and +2 [27, 46]. The Fourier-transformed EXAFS (FT-EXAFS) spectrum shows two distinct peaks at ca. 1.5 and 2.5 Å (Figure 1j), corresponding to Pd–O in PdO and Pd–Pd in metal Pd, respectively. Wavelet transform EXAFS (WT-EXAFS) reveals two intensity maxima ascribed to Pd–O and Pd–Pd scattering (Figure 1k and S9), confirming the coexistence of Pd and PdO_x domains. The characteristic Pd–Pd shell at ~3.0 Å, typically observed in PdO, is absent in both FT- and WT-EXAFS spectra, suggesting a reduced coordination number, likely due to oxygen deficiency in PdO_x [47]. Quantitative EXAFS fitting reveals an average Pd–Pd coordination number of ca. 4.2 (Table S2), much lower than that of Pd foil, indicating the formation of coordinatively unsaturated Pd sites (Figures S10). The Pd–O coordination number decreases from 4 in PdO to 2.3 in Pd@PdO_x/TiN, further corroborating the amorphous property of PdO_x [48]. Collectively, these findings reconfirm that the Pd@PdO_x NPs comprise Pd NPs with an amorphous PdO_x overlayer.

The CO₂ reduction performance of Pd@PdO_x/TiN was evaluated using H₂O as the electron donor under simulated sunlight. The catalyst with 0.8 wt.% Pd loading (Pd/PdO_x ratio of 1.96) shows the highest activity, generating CO and H₂ at rates of 200 μmol g⁻¹ h⁻¹ (4.00 μmol h⁻¹) and 47 μmol g⁻¹ h⁻¹ (0.94 μmol h⁻¹), respectively (Figure 2a), with a CO selectivity of 81%. Further increasing the Pd loading to 1.5 wt.%; however, leads to a decline in CO production. This is because excessive Pd likely introduces charge

recombination centers and partially blocks light absorption and surface active sites on TiN, thereby diminishing charge separation and interfacial transfer efficiency. This non-linear trend highlights the importance of optimizing cocatalyst loading to balance active-site density and charge dynamics. The apparent quantum efficiency (AQE_{CO}) of Pd@PdO_x/TiN is determined to be 0.15% under monochromatic irradiation at 365 nm. Such performance is favorably comparable to that of state-of-the-art catalysts for overall CO₂ reduction (Figure 2b and Table S3). However, bare TiN manifests only marginal yields of CO and H₂, underscoring the essential role of Pd species in activating CO₂ and promoting charge separation.

To demonstrate the cooperative roles of Pd and PdO_x in Pd@PdO_x/TiN, two control catalysts with comparable Pd loadings, but distinct Pd/PdO_x ratios (2.38 and 0.31) were synthesized (Figure S11). As shown in Figure 2c, the Pd-rich Pd@PdO_x/TiN catalyst (hereafter denoted as Pd/TiN) produces substantial H₂, resulting in a poor CO selectivity of 35%. In comparison, the PdO_x-rich Pd@PdO_x/TiN sample (denoted as: PdO_x/TiN) shows markedly lower formation of both CO and H₂, albeit with an increased CO selectivity of 92%. These results reveal that the coexistence of Pd and PdO_x species is vital for achieving efficient CO₂ activation and selective CO formation, stressing the synergistic yet competitive interplay between Pd and PdO_x in steering the photocatalytic CO₂ reduction pathway.

No products were detected in the absence of either light or catalyst, reflecting the photocatalytic nature of the reaction (Figure 2d). When the reaction was conducted in pure Ar, only H₂ was observed, excluding any contribution from carbonaceous contaminants. Isotopic labeling reactions using ¹³CO₂ produce

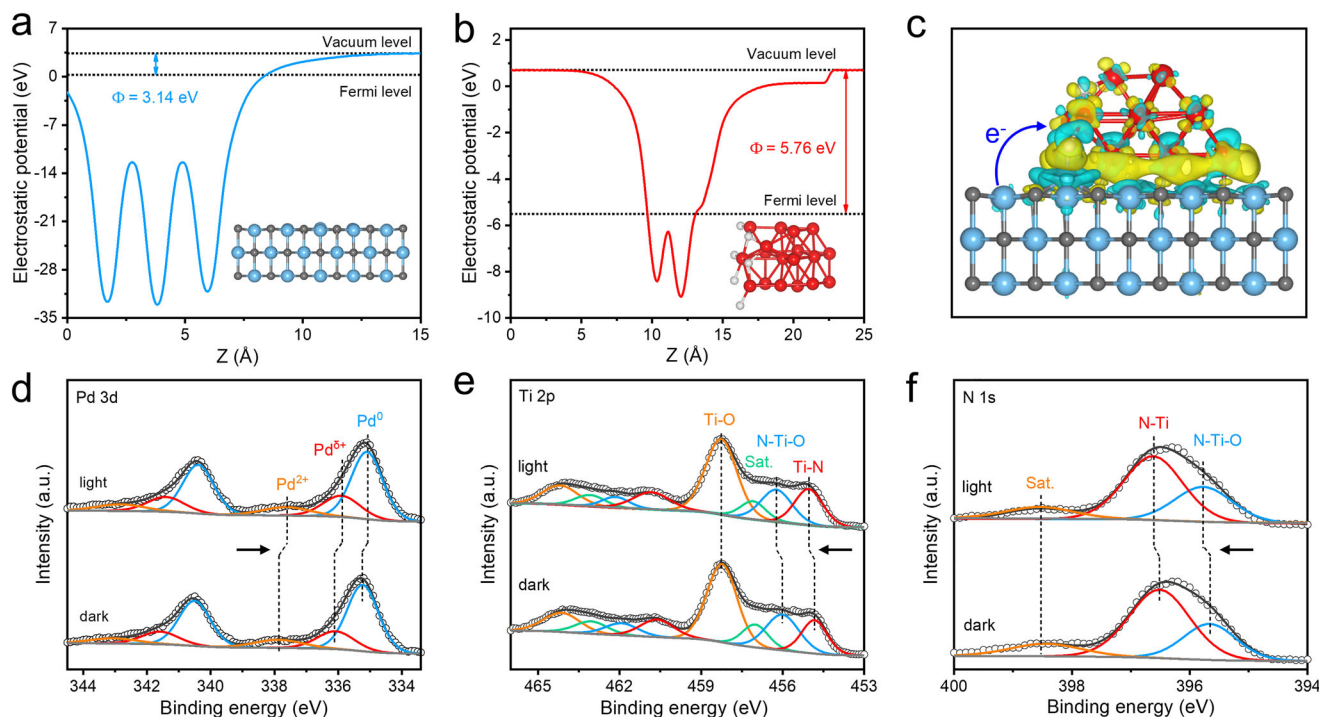


FIGURE 3 | The work functions and electrostatic potentials of (a) TiN and (b) Pd@PdO_x calculated by DFT calculations. (c) The differential charge density map of Pd@PdO_x/TiN, where the yellow and cyan electron clouds represent aggregation and outflow of electrons. The models of the calculated structure are listed, in which blue, black, red, and white balls represent Ti, N, Pd, and O, respectively. XPS spectra of (d) Pd 3d, (e) Ti 2p, and (f) N 1s of Pd@PdO_x/TiN.

¹³CO (Figure S12), unequivocally validating that CO₂ serves as the sole carbon source. Moreover, ¹⁸O-labeling experiments identify O₂ and H₂O₂ as oxidation products derived from H₂O (Figure S13). Furthermore, the electron-to-hole ratio was determined to be 1.03 (Figure S14), which is very close to the theoretical stoichiometric value. This quantitative balance reveals that the reduction and oxidation half-reactions are well matched and confirms that water serves as the hole-consuming species.

The activity of Pd@PdO_x/TiN varies with the wavelength of incident light (Figure 2e), confirming the light-driven nature of the reaction [49]. Strikingly, its activity decreases sharply under dark conditions, even at temperatures up to 380 °C, proving that photonic excitation, rather than light-induced heating, dominantly drives the CO₂ reduction process (Figure S15). Control experiments at 20 °C confirm that Pd@PdO_x/TiN remains active under illumination (Figure S15), further verifying that CO formation originates from a genuine photo-driven process. The enhanced performance at 280 °C is attributed to photothermal synergy, where photoinduced charge separation drives interfacial CO₂ activation while elevated temperature accelerates surface reaction kinetics and mass transport, collectively lowering the overall kinetic barriers.

The catalyst also parades decent durability, maintaining stable performance over five consecutive cycles (Figure 2f). During the stability tests, the total generation of CO/H₂ reaches 21.70 μmol, delivering a turnover number (TON) of 17.22 based on the total Pd species (Table S1). Post-reaction XPS analysis finds negligible changes in surface chemical states (Figure S16), substantiating the structural stability of Pd@PdO_x/TiN.

Afterward, a series of characterizations was conducted to explore the origin of the high activity of Pd@PdO_x/TiN. Upon incorporation of Pd@PdO_x, the composite displays intense and broad light absorption across the full solar spectrum (Figure S17). CO₂ adsorption isotherms disclose heightened CO₂ uptake of Pd@PdO_x/TiN compared to TiN (Figure S18a), verifying the porous feature of the cocatalyst. The CO₂-temperature-programmed desorption (CO₂-TPD) profiles present evident shifts of desorption peaks toward higher temperatures in weak and medium adsorption regions for Pd@PdO_x/TiN (Figure S18b). Importantly, an additional desorption emerges around 600 °C, indicative of newly formed strong basic sites that boost CO₂ activation. These results validate that Pd@PdO_x/TiN possesses improved CO₂ adsorption and activation ability, supporting its superior photocatalytic CO₂ reduction performance.

To clarify the charge separation and transfer behaviors, the work functions of Pd@PdO_x and TiN, as well as the spatial charge density difference of Pd@PdO_x/TiN, were determined by DFT calculations. Pd@PdO_x exhibits a greater work function (5.76 eV) than TiN (3.14 eV), implying the formation of a Schottky junction upon intimate contact (Figure 3a,b) [50–53]. Theoretically, electrons from TiN are expected to spontaneously transfer to Pd@PdO_x until their *E_f* equilibrate, resulting in upward band bending on the TiN side. The simulated interfacial charge density difference visually approves electron migration from TiN to Pd@PdO_x (Figures 3c and S19) [54, 55]. Under light irradiation, photoexcited electrons in TiN can migrate across the Schottky interface to Pd@PdO_x, where they accumulate to participate in CO₂ reduction.

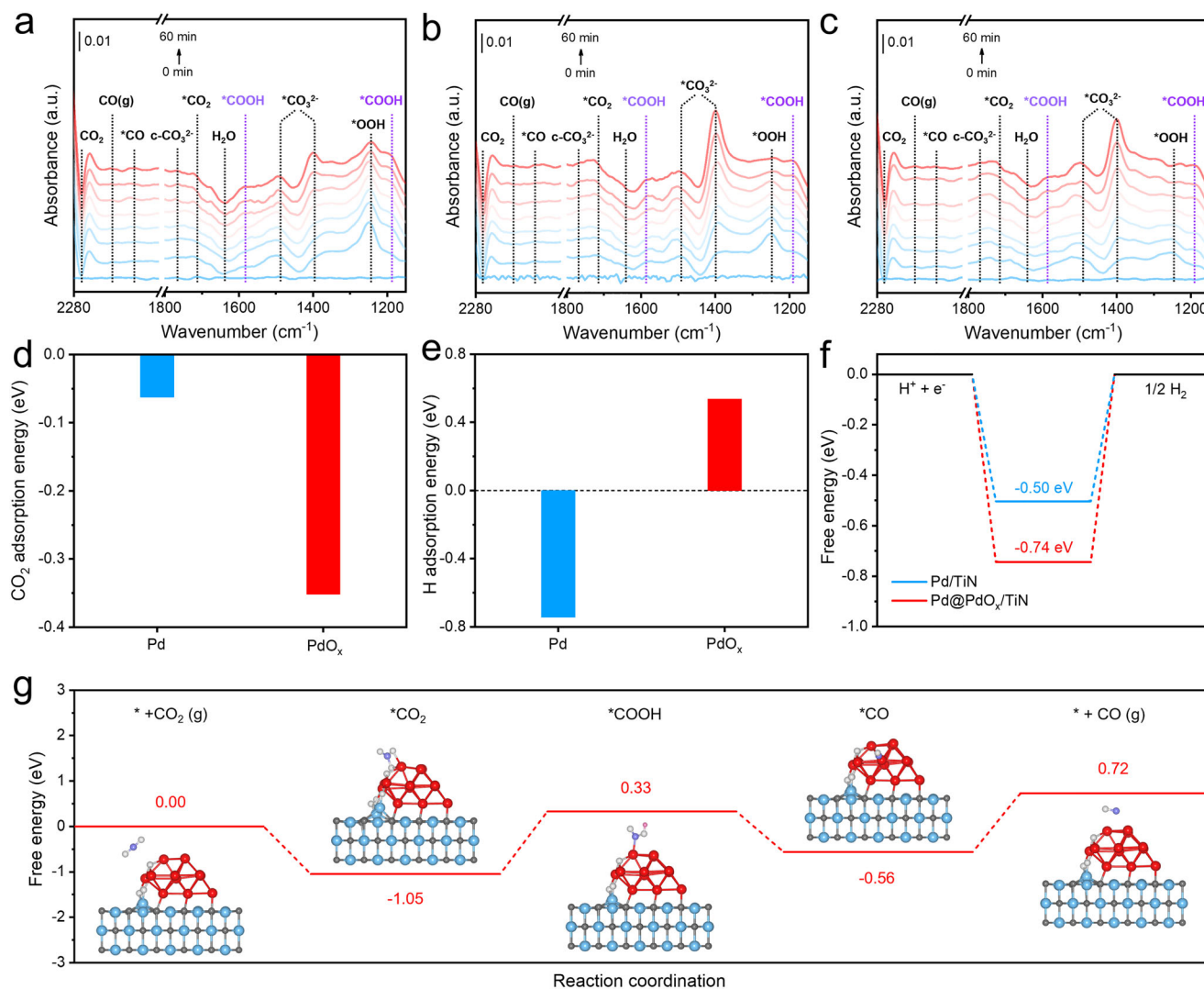


FIGURE 4 | In situ DRIFTS spectra of (a) Pd@PdO_x/TiN, (b) Pd/TiN, and (c) PdO_x/TiN in a CO₂ atmosphere. (d) CO₂ adsorption energy and (e) H adsorption energy of different sites in Pd@PdO_x/TiN. (f) Gibbs free energy of hydrogen evolution on the Pd site in Pd/TiN and Pd@PdO_x/TiN. (g) Gibbs free energy diagrams of CO₂-to-CO conversion pathways over Pd@PdO_x/TiN. The models of the calculated structure are listed, in which blue, black, red, white, and purple balls represent Ti, N, Pd, O, and C, respectively.

In situ irradiated XPS (ISI-XPS) was employed to demonstrate the migration direction of charge carriers in Pd@PdO_x/TiN. Upon illumination, the Pd 3d binding energies of all Pd species shift negatively (Figure 3d), indicating electron gathering on Pd@PdO_x. Simultaneously, the binding energies of N–Ti and N–Ti–O bonds in both the Ti 2p and N 1s spectra shift positively (Figure 3e, f), suggesting hole accumulation on the TiN surface. These findings are consistent with the DFT-derived charge redistribution.

The charge separation and transfer dynamics were examined. Pd@PdO_x/TiN delivers a much higher photocurrent response than TiN (Figure S20a), evidencing its accelerated charge transport [56]. Electrochemical impedance spectroscopy (EIS) reveals a smaller semicircular radius for Pd@PdO_x/TiN (Figure S20b), indicating reduced interfacial charge-transfer resistance [57]. In addition, the photoluminescence (PL) intensity of TiN is quenched upon Pd@PdO_x decoration (Figure S20c), implying effective suppression of carrier recombination [58, 59]. These

results collectively underline that the Schottky barrier formed at the interface of Pd@PdO_x and TiN promotes directional electron movement to the cocatalyst, inhibits electron backflow and enhances charge separation, thus facilitating photocatalytic CO₂ reduction.

To monitor the surface reaction intermediates, in situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) measurements were executed. The spectra captured in the dark indicate that Pd@PdO_x/TiN, Pd/TiN, and PdO_x/TiN all effectively adsorb CO₂ (Figure S21). Upon irradiation, inverted peaks at 1620 and 2258 cm⁻¹ gradually intensify (Figure 4a), corresponding to surface-adsorbed H₂O and CO₂, revealing their simultaneous adsorption and activation on Pd@PdO_x/TiN. As irradiation proceeds, new features assigned to activated CO₂ species (CO₂^{*}, 1724 cm⁻¹), chelating carbonate (c-CO₃²⁻, 1764 cm⁻¹), and monodentate carbonate (m-CO₃²⁻, 1398 and 1496 cm⁻¹) appear, suggesting progressive CO₂ activation and conversion [60–62]. Remarkably, distinct bands at 1180 and 1580 cm⁻¹ attributed to

*COOH intermediates are clearly detected [63]. These *COOH species then evolve into adsorbed CO (*CO, 2120 cm⁻¹) and gaseous CO (CO(g), 2178 cm⁻¹) [9]. Meanwhile, a vibration corresponding to *OOH species (1243 cm⁻¹) is also observed, which is an essential intermediate in the H₂O oxidation reaction [64].

Under identical conditions, Pd/TiN displays the strongest *CO₂ and m-CO₃²⁻ signals, but the weakest *COOH formation (Figures 4b and S22a–c), suggesting that CO₂-to-*COOH conversion is kinetically repressed. In contrast, PdO_x/TiN shows moderate CO₂ activation yet minimal *OOH formation (Figures 4c and S22d), signifying that although PdO_x/TiN facilitates CO₂-to-*COOH conversion more effectively than Pd/TiN, its overall redox pathway is limited by sluggish *H supply and transfer kinetics.

Complementary CO₂-TPD and linear sweep voltammetry (LSV) tests reveal that Pd chiefly promotes the H₂ evolution reaction, whereas PdO_x reinforces CO₂ adsorption and activation (Figures S23 and S24) [65, 66]. Taken together with photocatalytic performance, these results suggest a cooperative mechanism in Pd@PdO_x/TiN: Pd domains act as *H reservoirs, while PdO_x sites adsorb and activate CO₂. The hydrogenation of CO₂ to *COOH occurs via interfacial *H transfer from Pd to PdO_x, establishing a synergistic Pd–PdO_x coupling on TiN that enables efficient and selective CO₂ reduction.

DFT calculations provide further mechanistic insights into the adsorption, activation, and conversion process of CO₂ reduction by H₂O. The preferential adsorption sites of CO₂ and H were first estimated by comparing their adsorption energies on Pd and PdO_x at the heterointerface. PdO_x exhibits a more negative CO₂ adsorption energy (−0.35 eV) than Pd (−0.06 eV), indicating its stronger affinity toward CO₂ molecules (Figures 4d and S25). Conversely, the H adsorption energy on PdO_x (0.54 eV) is considerably higher than that on Pd (−0.74 eV), suggesting that Pd acts as a proton collector (Figures 4e and S26). The positive adsorption energy on PdO_x reflects weak H binding, thereby enabling *H delivery to assist CO₂ reduction. Also, the Gibbs free energy of hydrogen adsorption (ΔG_H) was assessed to describe the H₂ evolution tendency. The ΔG_H value on Pd/TiN is closer to zero than that on Pd@PdO_x/TiN (Figures 4f, S27, and S28), implying that Pd serves as the primary site for proton reduction and H₂ desorption, whereas H₂ evolution is relatively suppressed on Pd@PdO_x/TiN—beneficial for directing *H species for CO₂ reduction.

*COOH is identified as the key intermediate in the CO₂-to-CO conversion pathway. On PdO_x/TiN, *COOH formation requires a high barrier of 1.62 eV, constituting the potential-determining step (Figures S29, S30). In contrast, the barrier reduces to 1.38 eV on Pd@PdO_x/TiN, evidencing that Pd–PdO_x interfacial synergy significantly accelerates *COOH formation (Figure 4g). The subsequent conversion of *COOH to *CO on Pd@PdO_x/TiN is exergonic (−0.89 eV), indicating a thermodynamically favorable and spontaneous process. Furthermore, CO desorption on Pd@PdO_x/TiN (ΔE = 1.29 eV) is energetically easier than on PdO_x/TiN (ΔE = 1.39 eV), confirming the superior CO release capability of the Pd–PdO_x interface. These theoretical results corroborate the experimental observations, emphasizing that Pd and PdO_x species act cooperatively—Pd domains serve as

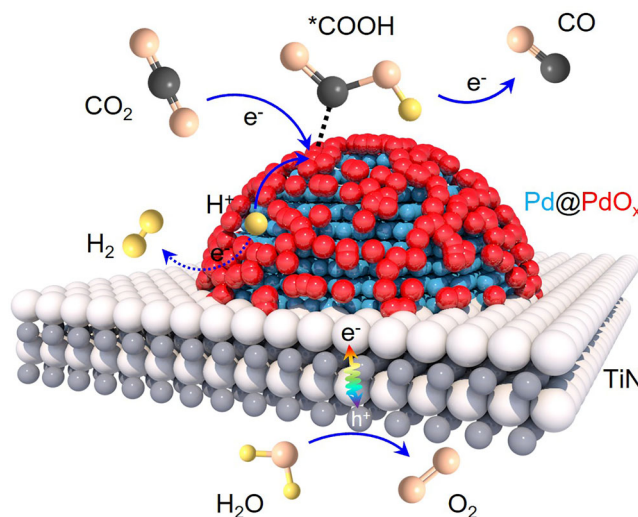


FIGURE 5 | Illustration of the proposed reaction processes of photocatalytic CO₂ reduction with H₂O over Pd@PdO_x/TiN.

*H suppliers, while PdO_x sites anchor and activate CO₂. The interfacial *H transfer between Pd and PdO_x lowers the potential barrier for *COOH formation, accelerating CO₂-to-CO conversion on Pd@PdO_x/TiN. Additionally, the spatial separation of proton collection and CO₂ activation in Pd@PdO_x suppresses *CO over-hydrogenation, which, together with the thermodynamically favorable formation and facile desorption of *CO, ensures high CO selectivity and effectively prevents CH₄ formation.

Based on combined experimental and theoretical findings, the photocatalytic CO₂ reduction pathway over Pd@PdO_x/TiN hybrid is proposed (Figure 5). Upon light irradiation, the metallic TiN photocatalyst is motivated to produce charge carriers. The photoexcited electrons are then transferred to the Pd@PdO_x cocatalyst through the Schottky interface, while the holes remain on the TiN support. Within the dual-function cocatalyst, PdO_x species serve as the active sites for CO₂ adsorption and activation, whereas adjacent Pd sites act as proton collectors. The interfacial transfer of protons from Pd to PdO_x augments the formation of the key *COOH intermediate, promoting CO₂-to-CO reduction while concurrently suppressing the competing hydrogen evolution reaction. This synergistic reaction pathway empowers selective CO generation with enhanced overall efficiency.

3 | Conclusion

In summary, a dual-functional Pd@PdO_x cocatalyst anchored on metallic TiN enables efficient and selective overall CO₂-to-CO conversion under full-spectrum irradiation. The well-defined Pd–PdO_x heterointerface facilitates rapid charge separation and directional electron transfer from TiN to the cocatalyst, while spatially coordinating proton management and CO₂ activation. Metal Pd domains function as a proton collector, while adjacent PdO_x sites intensify CO₂ adsorption and activation. Directional proton transfer across the interface lowers the energy barrier for *COOH formation and suppresses the competing hydrogen evolution reaction, thereby enhancing CO selectivity. This work reveals that surface oxide species often considered incidental

oxidation states—serve as structural determinants of interfacial charge and proton dynamics. By mediating proton-coupled electron transfer at metal/oxide boundaries, surface oxides actively govern reaction-pathway selection in CO₂ photoreduction. These insights provide a mechanistic foundation for rationally engineering metal cocatalysts and metal/oxide heterointerfaces in artificial photosynthesis.

Author Contributions

Bifang Li: conceptualization, investigation, funding acquisition, writing – review & editing, and visualization. **Ziyi Sun:** investigation, funding acquisition, writing – original draft, visualization, project administration, formal analysis, and software. **Bo Su:** investigation, visualization. **Xiahui Lin:** resources, supervision, and data curation. **Wandong Xing:** project administration, formal analysis, and validation. **Yuanfu Ren:** writing – original draft, writing – review & editing, and project administration. **Kunlong Liu:** funding acquisition, formal analysis, software, and project administration. **Yidong Hou:** writing – review & editing, visualization. **Xue Feng Lu:** visualization, project administration. **Lihua Lin:** formal analysis, funding acquisition. **Masakazu Anpo:** formal analysis, funding acquisition. **Huabin Zhang:** investigation, funding acquisition, and writing – original draft. **Sibo Wang:** investigation, funding acquisition, validation, formal analysis, visualization, writing – original draft, and writing – review & editing.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (22372035, 22302039, 22432003, and 52201006), the Science Foundation of the Fujian Province (2022L3084), the III Project (D16008) and Center of Excellence for Renewable Energy and Storage Technologies under award number 5937.

Conflicts of Interest

The authors declare no conflicts of interests.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request

References

- W. Wang, W. Zhang, C. Deng, H. Sheng, and J. Zhao, “Accelerated Photocatalytic Carbon Dioxide Reduction and Water Oxidation under Spatial Synergy,” *Angewandte Chemie International Edition* 63 (2024): e202317969, <https://doi.org/10.1002/anie.202317969>.
- H. Ou, S. Ning, P. Zhu, et al., “Carbon Nitride Photocatalysts With Integrated Oxidation and Reduction Atomic Active Centers for Improved CO₂ Conversion,” *Angewandte Chemie International Edition* 61 (2022): e202206579, <https://doi.org/10.1002/anie.202206579>.
- X. Li, L. Li, G. Chen, et al., “Accessing Parity-Forbidden *d-d* Transitions for Photocatalytic CO₂ Reduction Driven by Infrared Light,” *Nature Communications* 14 (2023): 4034, <https://doi.org/10.1038/s41467-023-39666-0>.
- Y. Bao, S. Du, K. Shibata, et al., “Layered β -ZrNBr Nitro-Halide as Multifunctional Photocatalyst for Water Splitting and CO₂ Reduction,” *Angewandte Chemie International Edition* 62 (2023): e202214273, <https://doi.org/10.1002/anie.202214273>.
- K. Iizuka, T. Wato, Y. Miseki, K. Saito, and A. Kudo, “Photocatalytic Reduction of Carbon Dioxide Over Ag Cocatalyst-Loaded ALa₄Ti₄O₁₅ (A = Ca, Sr, and Ba) Using Water as a Reducing Reagent,” *Journal of*

the American Chemical Society 133 (2011): 20863–20868, <https://doi.org/10.1021/ja207586e>.

- A. Iwase, S. Yoshino, T. Takayama, Y. H. Ng, R. Amal, and A. Kudo, “Water Splitting and CO₂ Reduction Under Visible Light Irradiation Using Z-Scheme Systems Consisting of Metal Sulfides, CoOx-Loaded BiVO₄, and a Reduced Graphene Oxide Electron Mediator,” *Journal of the American Chemical Society* 138 (2016): 10260–10264, <https://doi.org/10.1021/jacs.6b05304>.
- K. Teramura, S. Iguchi, Y. Mizuno, T. Shishido, and T. Tanaka, “Photocatalytic Conversion of CO₂ in Water Over Layered Double Hydroxides,” *Angewandte Chemie International Edition* 51 (2012): 8008–8011, <https://doi.org/10.1002/anie.201201847>.
- R. Li, J. Hu, M. Deng, et al., “Integration of an Inorganic Semiconductor With a Metal–Organic Framework: A Platform for Enhanced Gaseous Photocatalytic Reactions,” *Advanced Materials* 26 (2014): 4783–4788, <https://doi.org/10.1002/adma.201400428>.
- S. Hu, P. Qiao, X. Yi, et al., “Selective Photocatalytic Reduction of CO₂ to CO Mediated by Silver Single Atoms Anchored on Tubular Carbon Nitride,” *Angewandte Chemie International Edition* 62 (2023): e202304585, <https://doi.org/10.1002/anie.202304585>.
- X. Shi, Y. Huang, Y. Bo, et al., “Highly Selective Photocatalytic CO₂ Methanation With Water Vapor on Single-Atom Platinum-Decorated Defective Carbon Nitride,” *Angewandte Chemie International Edition* 61 (2022): e202203063, <https://doi.org/10.1002/ange.202203063>.
- C. Ding, X. Lu, B. Tao, et al., “Interlayer Spacing Regulation by Single-Atom Indium^{δ+} –N₄ on Carbon Nitride for Boosting CO₂/CO Photo-Conversion,” *Advanced Functional Materials* 33 (2023): 2302824, <https://doi.org/10.1002/adfm.202302824>.
- Z. Li, P. Wang, C. Ren, et al., “Modulating the Selectivity of CO₂ Photoreduction by Regulating the Location of PtCu in a UiO-66@ZnIn₂S₄ Core–Shell Nanoreactor,” *ACS Catalysis* 15 (2025): 828–840, <https://doi.org/10.1021/acscatal.4c05717>.
- M. Zhou, H. Wang, R. Liu, et al., “Construction of Frustrated Lewis Pairs in Poly(heptazine Imide) Nanosheets via Hydrogen Bonds for Boosting CO₂ Photoreduction,” *Angewandte Chemie International Edition* 63 (2024): e202407468, <https://doi.org/10.1002/anie.202407468>.
- C. Guan, X. Yue, Y. Liao, and Q. Xiang, “Structurally Asymmetric Ni–O–Mn Node in Metal–Organic Layers on Carbon Nitride Support for CO₂ Photoreduction,” *Angewandte Chemie International Edition* 64 (2025): e202415538, <https://doi.org/10.1002/anie.202415538>.
- Y.-S. Xia, L. Zhang, J.-N. Lu, et al., “A Triple Tandem Reaction for the Upcycling of Products From Poorly Selective CO₂ Photoreduction Systems,” *Nature Synthesis* 3 (2024): 406–418, <https://doi.org/10.1038/s44160-023-00458-5>.
- J. Li, H. Huang, W. Xue, et al., “Self-Adaptive Dual-Metal-Site Pairs in Metal–Organic Frameworks for Selective CO₂ Photoreduction to CH₄,” *Nature Catalysis* 4 (2021): 719–729, <https://doi.org/10.1038/s41929-021-00665-3>.
- H. Qiu, A. Yamamoto, and H. Yoshida, “Surface Gallium Oxide Hydroxide Species Adsorbing Carbon Dioxide to Enhance the Photocatalytic Activity of Silver-Loaded Calcium Titanate for Carbon Dioxide Reduction With Water,” *Sustain Energy Fuels* 8 (2024): 1287–1294, <https://doi.org/10.1039/D3SE01390A>.
- R. Pang, K. Teramura, H. Asakura, S. Hosokawa, and T. Tanaka, “Effect of Thickness of Chromium Hydroxide Layer on Ag Cocatalyst Surface for Highly Selective Photocatalytic Conversion of CO₂ by H₂O,” *ACS Sustainable Chemistry & Engineering* 7 (2019): 2083–2090, <https://doi.org/10.1021/acssuschemeng.8b04665>.
- W. Shangguan, G. Li, S. Gui, et al., “Restructuring at Au/AlOOH Interface Enables Enhanced CO₂ Photoreduction by Synergistically Optimizing Charge Separation and H₂O Activation,” *Advanced Functional Materials* 35 (2025): 2501285, <https://doi.org/10.1002/adfm.202501285>.
- Y. Ma, X. Yi, S. Wang, et al., “Selective Photocatalytic CO₂ Reduction in Aerobic Environment by Microporous Pd-Porphyrin-Based Polymers

- Coated Hollow TiO₂,” *Nature Communications* 13 (2022): 1400, <https://doi.org/10.1038/s41467-022-29102-0>.
21. X. Ma, Q. Chen, C. Han, et al., “Synergistic Optimization of Morphology and Vacancies on Diatomic Rhodium Catalysts Dispersed on Carbon Nitride for Efficient Photocatalytic Reduction of CO₂,” *Advanced Functional Materials* 34 (2024): 2307733, <https://doi.org/10.1002/adfm.202307733>.
 22. F. Mo, Q. Zhou, W. Xue, et al., “The Optimized Catalytic Performance of Single-Atom Catalysts by Incorporating Atomic Clusters or Nanoparticles: In-Depth Understanding on Their Synergisms,” *Advanced Energy Materials* 13 (2023): 2301711, <https://doi.org/10.1002/aenm.202301711>.
 23. R. You, M. Ma, Z. Wang, et al., “Unveiling Water-Vapor-Promoted Oxidation of Palladium Nanoparticles via Atomic-Scale Transmission Electron Microscopy at Atmospheric Pressure,” *Journal of the American Chemical Society* 147 (2025): 28035–28043, <https://doi.org/10.1021/jacs.5c07765>.
 24. Z. Hu, J. Zhu, R. Chen, et al., “High-Rate and Selective C₂H₆-to-C₂H₄ Photodehydrogenation Enabled by Partially Oxidized Pd^{δ+} Species Anchored on ZnO Nanosheets Under Mild Conditions,” *Journal of the American Chemical Society* 146 (2024): 16490–16498, <https://doi.org/10.1021/jacs.4c02827>.
 25. Z.-Z. Wu, F.-Y. Gao, and M.-R. Gao, “Regulating the Oxidation state of Nanomaterials for Electrocatalytic CO₂ Reduction,” *Energy & Environmental Science* 14 (2021): 1121–1139, <https://doi.org/10.1039/D0EE02747B>.
 26. T. P. Araújo, J. Morales-Vidal, G. Giannakakis, et al., “Reaction-Induced Metal-Metal Oxide Interactions in Pd-In₂O₃/ZrO₂ Catalysts Drive Selective and Stable CO₂ Hydrogenation to Methanol,” *Angewandte Chemie International Edition* 62 (2023): e202306563, <https://doi.org/10.1002/anie.202306563>.
 27. P. Liu, Z. Huang, X. Gao, et al., “Synergy Between Palladium Single Atoms and Nanoparticles via Hydrogen Spillover for Enhancing CO₂ Photoreduction to CH₄,” *Advanced Materials* 34 (2022): 2200057, <https://doi.org/10.1002/adma.202200057>.
 28. S. Liu, Y. Li, X. Yu, et al., “Tuning Crystal-Phase of Bimetallic Single-Nanoparticle for Catalytic Hydrogenation,” *Nature Communications* 13 (2022): 4559, <https://doi.org/10.1038/s41467-022-32274-4>.
 29. S. Latiyan, D. K. Yadav, M. R. Das, A. Singh, P. Rajput, and S. Deka, “Maximized Synergistic Interaction of Pd Single Atoms With CuCo₂S₄ Nanosheets Catalysts for High-Efficiency Photocatalytic H₂ Production and O₂ Reduction via Enhanced Charge Separation and Transfer,” *Advanced Functional Materials* 35 (2025): 202507783, <https://doi.org/10.1002/adfm.202507783>.
 30. H. Zhang, A. Yohannes, H. Zhao, et al., “Photocatalytic Asymmetric C-C Coupling for CO₂ Reduction on Dynamically Reconstructed Ru^{δ+}-O/Ru⁰-O Sites,” *Nature Communications* 16 (2025): 534, <https://doi.org/10.1038/s41467-025-55885-z>.
 31. Y. Wu, Z. Li, Q. Chen, Z. Lou, G. Wang, and J. Mao, “Cooperative Atomic Palladium Site and Island-Distributed S-Scheme Heterostructure for Photocatalytic C₂H₆ Production,” *ACS Catalysis* 15 (2025): 3558–3569, <https://doi.org/10.1021/acscatal.4c07797>.
 32. B. Su, Y. Kong, S. Wang, et al., “Hydroxyl-Bonded Ru on Metallic TiN Surface Catalyzing CO₂ Reduction With H₂O by Infrared Light,” *Journal of the American Chemical Society* 145 (2023): 27415–27423, <https://doi.org/10.1021/jacs.3c08311>.
 33. J. Zhao, R. Urrego-Ortiz, N. Liao, F. Calle-Vallejo, and J. Luo, “Rationally Designed Ru Catalysts Supported on TiN for Highly Efficient and Stable Hydrogen Evolution in Alkaline Conditions,” *Nature Communications* 15 (2024): 6391, <https://doi.org/10.1038/s41467-024-50691-5>.
 34. X. Wu, W. Zhang, J. Li, Q. Xiang, Z. Liu, and B. Liu, “Identification of the Active Sites on Metallic MoO_{2-x} Nano-Sea-Urchin for Atmospheric CO₂ Photoreduction Under UV, Visible, and Near-Infrared Light Illumination,” *Angewandte Chemie International Edition* 62 (2022): e202213124, <https://doi.org/10.1002/anie.202213124>.
 35. Z. Hu, G. Liu, X. Chen, Z. Shen, and J. C. Yu, “Enhancing Charge Separation in Metallic Photocatalysts: A Case Study of the Conducting Molybdenum Dioxide,” *Advanced Functional Materials* 26 (2016): 4445–4455, <https://doi.org/10.1002/adfm.201600239>.
 36. J. Xu, Z. Ju, W. Zhang, et al., “Efficient Infrared-Light-Driven CO₂ Reduction over Ultrathin Metallic Ni-Doped CoS₂ Nanosheets,” *Angewandte Chemie International Edition* 60 (2021): 8705–8709, <https://doi.org/10.1002/anie.202017041>.
 37. M. Liu, G. Zhang, X. Liang, et al., “Rh/Cr₂O₃ and CoO_x Cocatalysts for Efficient Photocatalytic Water Splitting by Poly (Triazine Imide) Crystals,” *Angewandte Chemie International Edition* 62 (2023): e202304694, <https://doi.org/10.1002/anie.202304694>.
 38. F. Liu, J. Deng, B. Su, et al., “Poly(triazine imide) Crystals for Efficient CO₂ Photoreduction: Surface Pyridine Nitrogen Dominates the Performance,” *ACS Catalysis* 15 (2025): 1018–1026, <https://doi.org/10.1021/acscatal.4c06685>.
 39. T. Wang, L. Zhang, J. Wu, et al., “Few-Layer Fullerene Network for Photocatalytic Pure Water Splitting Into H₂ and H₂O₂,” *Angewandte Chemie International Edition* 62 (2023): e202311352, <https://doi.org/10.1002/anie.202311352>.
 40. J. Kang, X. Yang, Q. Hu, Z. Cai, L.-M. Liu, and L. Guo, “Recent Progress of Amorphous Nanomaterials,” *Chemical Reviews* 123 (2023): 8859–8941, <https://doi.org/10.1021/acs.chemrev.3c00229>.
 41. H. Li, J. Yu, Y. Sui, et al., “Constructing Palladium-Based Crystalline@Amorphous Core-Shell Heterojunctions for Efficient Formic Acid Oxidation,” *Advanced Science* 12 (2025): 202504469, <https://doi.org/10.1002/advs.202504469>.
 42. X. Jin, T. Lee, W. Tamakloe, et al., “In Situ Defect Engineering Route to Optimize the Cationic Redox Activity of Layered Double Hydroxide Nanosheet via Strong Electronic Coupling With Holey Substrate,” *Advanced Science* 9 (2022): 2103368, <https://doi.org/10.1002/advs.202103368>.
 43. C. Mao, J. Wang, Y. Zou, et al., “Photochemical Acceleration of Ammonia Production by Pt₁-Pt_n-TiN Reduction and N₂ Activation,” *Journal of the American Chemical Society* 145 (2023): 13134–13146, <https://doi.org/10.1021/jacs.3c01947>.
 44. S. Yue, C. S. Praveen, A. Klyushin, et al., “Redox Dynamics and Surface Structures of an Active Palladium Catalyst During Methane Oxidation,” *Nature Communications* 15 (2024): 4678, <https://doi.org/10.1038/s41467-024-49134-y>.
 45. H. Xiong, D. Kunwar, D. Jiang, et al., “Engineering Catalyst Supports to Stabilize PdO_x Two-dimensional Rafts for Water-Tolerant Methane Oxidation,” *Nature Catalysis* 4 (2021): 830–839, <https://doi.org/10.1038/s41929-021-00680-4>.
 46. K. Zheng, S. Liu, J. Zhu, et al., “High-Rate CO₂-to-CH₄ Photoreduction by Dual-Proton Hydrogenation Pathway Over Pd-Anchored Oxygen-Deficient ZnO Nanosheets,” *Angewandte Chemie International Edition* 64 (2025): e202508259, <https://doi.org/10.1002/anie.202508259>.
 47. B. Wu, X. Liu, P. Liu, et al., “Synthesis of Amorphous Metal Oxides via a Crystalline to Amorphous Phase Transition Strategy,” *Nature Synthesis* 4 (2025): 370–379, <https://doi.org/10.1038/s44160-024-00704-4>.
 48. V. Muravev, J. F. M. Simons, A. Parastae, et al., “Operando Spectroscopy Unveils the Catalytic Role of Different Palladium Oxidation States in CO Oxidation on Pd/CeO₂ Catalysts,” *Angewandte Chemie International Edition* 61 (2022): e202200434, <https://doi.org/10.1002/anie.202200434>.
 49. X. Xinyu, J. Zhou, G. Meiyan, et al., “Redox-Pathway Steering Enables Photocatalytic Valorization of CO₂ and 1-Phenylethanol With Record Quantum Efficiency,” *Science Bulletin* 71 (2026): 1695–1702, <https://doi.org/10.1016/j.scib.2026.02.031>.

50. X. Zhang, F. Wu, G. Li, et al., "Dual Electric Field Coupling With Tunable Schottky Barrier Synergistically Regulating Electronic Configuration in CoP@Ni-CdS Heterojunction For Efficient Photocatalytic H₂ Evolution," *Advanced Functional Materials* 35 (2024): 202412527, <https://doi.org/10.1002/adfm.202412527>.
51. G. Zuo, Y. Wang, W. L. Teo, et al., "Ultrathin ZnIn₂S₄ Nanosheets Anchored on Ti₃C₂T_x MXene for Photocatalytic H₂ Evolution," *Angewandte Chemie International Edition* 59 (2020): 11287–11292, <https://doi.org/10.1002/anie.202002136>.
52. A. Wang, S. Wu, J. Dong, et al., "Interfacial Facet Engineering on the Schottky Barrier Between Plasmonic Au and TiO₂ in Boosting the Photocatalytic CO₂ Reduction Under Ultraviolet and Visible Light Irradiation," *Chemical Engineering Journal* 404 (2021): 127145, <https://doi.org/10.1016/j.cej.2020.127145>.
53. Y. Zheng, S. Ling, Y. Lv, et al., "Enhanced Adsorption Mediates Efficient CO₂ Reduction Over a Ru/TiO_{2-x} Schottky Heterojunction," *Energy & Fuels* 38 (2024): 18909–18917, <https://doi.org/10.1021/acs.energyfuels.4c03598>.
54. Y. Wu, D. Yang, Y. Zhang, et al., "Integrated Unit-Cell-thin MXene and Schottky Electric Field Into Piezo-Photocatalyst for Enhanced Photocatalytic Separation and Hydrogen Evolution," *Chemical Engineering Journal* 439 (2022): 135640, <https://doi.org/10.1016/j.cej.2022.135640>.
55. Y. Xu, Z. Zhou, P. Yu, and Y. Wang, "Al-doped CuS/ZnIn₂S₄ Forming Schottky Junction Induced Electron Trap Centers for Photocatalytic Overall Water Splitting," *Chemical Engineering Journal* 470 (2023): 144275, <https://doi.org/10.1016/j.cej.2023.144275>.
56. L. Huang, B. Li, B. Su, et al., "Fabrication of Hierarchical Co₃O₄@CdIn₂S₄ p-n Heterojunction Photocatalysts for Improved CO₂ Reduction With Visible Light," *Journal of Materials Chemistry A* 8 (2020): 7177–7183, <https://doi.org/10.1039/D0TA01817A>.
57. B. Li, W. Wang, J. Zhao, et al., "All-Solid-State Direct Z-scheme NiTiO₃/Cd_{0.5}Zn_{0.5}S Heterostructures for Photocatalytic Hydrogen Evolution With Visible Light," *Journal of Materials Chemistry A* 9 (2021): 10270–10276, <https://doi.org/10.1039/D1TA01220G>.
58. Y. Liu, W. Xue, X. Liu, et al., "Ultrafine Pt Nanoparticles on Defective Tungsten Oxide for Photocatalytic Ethylene Synthesis," *Small* 20 (2024): 2402004, <https://doi.org/10.1002/smll.202402004>.
59. T. Yang, H. Yang, B. Su, et al., "Regulating Product Selectivity in Photothermal CO₂ Conversion Through Ni Loading on Metallic MoO₃," *ACS Sustainable Chemistry & Engineering* 14 (2026): 827–836, <https://doi.org/10.1021/acssuschemeng.5c12251>.
60. L. Ran, Z. Li, B. Ran, et al., "Engineering Single-Atom Active Sites on Covalent Organic Frameworks for Boosting CO₂ Photoreduction," *Journal of the American Chemical Society* 144 (2022): 17097–17109, <https://doi.org/10.1021/jacs.2c06920>.
61. K. Zhu, Q. Zhu, M. Jiang, et al., "Modulating Ti_{2g} Orbital Occupancy in a Cu/TiO₂ Composite for Selective Photocatalytic CO₂ Reduction to CO," *Angewandte Chemie International Edition* 61 (2022): e202207600, <https://doi.org/10.1002/anie.202207600>.
62. L. Liu, H. Huang, Z. Chen, et al., "Synergistic Polarization Engineering on Bulk and Surface for Boosting CO₂ Photoreduction," *Angewandte Chemie International Edition* 60 (2021): 18303–18308, <https://doi.org/10.1002/anie.202106310>.
63. W. Shangguan, Q. Liu, Y. Wang, et al., "Molecular-Level Insight Into Photocatalytic CO₂ Reduction With H₂O Over Au Nanoparticles by Interband Transitions," *Nature Communications* 13 (2022): 3894, <https://doi.org/10.1038/s41467-022-31474-2>.
64. C. Feng, J. Luo, C. Chen, et al., "Cooperative Tungsten Centers in Polymeric Carbon Nitride for Efficient Overall Photosynthesis of Hydrogen Peroxide," *Energy & Environmental Science* 17 (2024): 1520–1530, <https://doi.org/10.1039/D3EE03032F>.
65. B. Su, S. Wang, W. Xing, et al., "Synergistic Ru Species on Poly(heptazine imide) Enabling Efficient Photocatalytic CO₂ Reduction With H₂O Beyond 800 nm," *Angewandte Chemie International Edition* 64 (2025): e202505453, <https://doi.org/10.1002/anie.202505453>.
66. B. Su, K. Tang, J. Cai, et al., "Electron-Deficient Ru^{δ+} Sites Coupled With Lattice Oxygen at Ru/BaTiO₃ Interfaces Enable Efficient Photothermal Methane Dry Reforming," *ACS Catalysis* 16 (2026): 5208–5217, <https://doi.org/10.1021/acscatal.6c00807>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: anie72396-sup-0001-SupMat.docx.